PATN Wishlist

May 26, 2026

1. Better 3d graphics
2. Implement Constancy and fidelity for 1/0 variables. Constancy is the % of objects within a group it occurs in). Fidelity is the % degree to which a variable is in one group (#in/#out).
3. Algorithm for determining an optimal subset of x objects from ordination (coordinates). Use SSH Lower Symmetric Matrix (LSM). This is Euclidean distance. Take two objects furthest apart as seeds then iterate to find the object that is furthest from selected objects until number of objects achieved. Simple and I think would be very effective for our current problem of minimal independent layers.
4. A simpler and more efficient ordination algorithm. These days, a simple non-metric MDS would suffice.
5. Import LSM with labels as raw data. Probably need to create dummy Data Table with labels and first column of LSM as dummy data.
6. Ability to mask in/out rows/columns of LSM (maybe to match masked rows of Data Table?)
7. Tool for subtracting two LSMs (absolute diff).
8. A mechanism for being able to retain a selection of objects from the Data Table (to identify on SSH plot). Often need to go back and forth from plot to table so selections lost.
9. ALOC: Allocate extrinsic rows to closest existing group.
10. Time links between pairs on ordination
11. Neural network algorithm.
12. Intergroup dissimilarity matrix - PCA - colour for ALOC
13. Generalised Linear Latent Variable Models (GLLVMs) are now widely regarded as the modern successor to classical ordination. They unify ordination and joint species distribution modelling (JSDMs). From AI:
    - Treat the site × species matrix as arising from a probabilistic model, not a distance matrix.
    - Handle presence/absence, counts, cover, biomass, etc., using appropriate likelihoods.
    - Estimate latent ecological gradients directly.
    - Allow species-specific responses to latent variables.
    - Provide uncertainty estimates for ordination axes.
    - Extend naturally to hierarchical and spatial structures.
14. Dirichlet-Process Ordination — simultaneous clustering + ordination:
    - Performs ordination and clustering simultaneously.
    - Automatically estimates the number of ecological communities (clusters).
    - Works directly with presence–absence data (your core use case).
    - Provides probabilistic statements about cluster membership.
    - Avoids pre-specifying the number of groups (unlike k-means or finite mixtures).
    - This is a major conceptual advance because it embeds ordination in a nonparametric Bayesian